10.4 How our energy is produced

This chapter will explain:

★ non-renewable and renewable energy sources
★ transforming the global energy mix to achieve a better balance between renewable and non-renewable sources of energy.

Non-renewable and renewable energy sources

Non-renewable energy sources are the fossil fuels (coal, oil, natural gas) and nuclear fuel. Eventually, these non-renewable resources could become completely exhausted. The burning of fossil fuels creates pollution and is the major source of greenhouse gas emissions. Climate change due to these emissions is the biggest environmental problem facing the planet.

▲ **Figure 10.26** Fuel station on the Amazon River, Brazil

Renewable energy resources are mainly forces of nature; these are sustainable and usually cause little or no pollution. Renewable energy includes hydroelectricity, biomass, wind, solar, geothermal, tidal energy and wave power. Hydroelectricity is the one renewable source of energy that is sometimes described as a traditional source of energy because water power has been used to generate electricity for over 100 years.

It is estimated that globally around 750 million people do not have access to electricity to light their homes and provide other services that most other people take for granted. In developing countries about 2.5 billion people rely on **fuelwood** (Figure 10.27) as their main source of energy. Wood and charcoal are collectively called fuelwood, and this accounts for just over half of global wood production. Fuelwood can be classed as either a renewable or a non-renewable source of energy, depending on whether or not the system of consumption allows the vegetation to renew itself to its previous density.

▲ **Figure 10.27** Woman carrying fuelwood from a local forest, Nepal

Transforming the global energy mix to achieve a better balance between renewable and non-renewable sources of energy

Fuelwood provides much of the energy needs for many countries in Africa. It is also the most important use of wood in Asia. Wood is likely to remain the main source of fuel for those in poverty around the world for the foreseeable future. The transition from fuelwood and animal dung to 'higher level' sources of energy, known as the **energy ladder**, occurs as part of the process of economic development. It requires a high level of investment, for example constructing an electricity transmission corridor (pylons and cables) to connect remote areas of a country that did not previously have electricity.

In LICs **energy poverty** has a big impact on people's lives and is a major obstacle to development. Energy poverty differs from **fuel poverty**. Fuel poverty is when people, usually in affluent countries, struggle to pay rising energy bills in harsh winters.

At present, non-renewable resources still dominate global energy supply. The challenge is to transform the global **energy mix** to achieve a better balance between renewable and non-renewable sources of energy. This is often referred to as the '**clean air transition**'.

Countries are eager to harness renewable energy resources in order to:

» reduce their reliance on domestic fossil fuel resources
» lower their reliance on costly fossil fuel imports
» improve their energy security
» cut greenhouse gas emissions.

The main drawback in the past to the new alternative energy sources was that they usually produced higher-cost electricity than traditional sources. However, the cost gap with wind and solar power closed rapidly in the first two decades of this century, and now these sources of energy are generally less costly than traditional energy, and without the climate-changing environmental costs.

Activities

1 Explain the difference between renewable and non-renewable energy sources.
2 What do you understand by the terms 'energy mix' and 'clean air transition'?
3 Briefly discuss two reasons why countries are keen to develop renewable sources of energy.